

➤ Customer Shopping Behavior Analysis — Integrated Python, SQL & Power BI Project

- **Executive Summary**

This project delivers a complete end-to-end data analysis system integrating Python, SQL (PostgreSQL), and Power BI to understand customer shopping behavior. Using a dataset of 3,900 transactions, the project covers data cleaning, feature engineering, database integration, business query execution, and development of an interactive dashboard.

- **Project Objectives**

1. Clean and preprocess customer shopping data using Python.
2. Create meaningful features such as age groups and purchase frequency.
3. Store cleaned data in PostgreSQL.
4. Write SQL queries for business questions.
5. Build an interactive Power BI dashboard.
6. Provide recommendations.

- **Dataset Summary**

- **Dataset:** customer_shopping_behavior_workbook.xlsx
- **Rows:** 3,900
- **Columns:** 18

- ❖ **Key Attributes:**

- Demographics (Age, Gender, Location)
- Purchase details (Item, Category, Amount)
- Behavior (Discounts, Previous Purchases, Ratings)
- Shipping Type

```
[6]: df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	1
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	1
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	1
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	1
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	1

- **Python Data Preparation & EDA**

Performed loading, cleaning, missing value treatment, feature engineering, and database integration using Python.

Initial Exploration: Used `df.info()` to check structure and `.describe()` for summary statistics.

```
[7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                        Non-Null Count  Dtype  
---  -
 0   Customer ID                  3900 non-null  int64  
 1   Age                          3900 non-null  int64  
 2   Gender                       3900 non-null  object  
 3   Item Purchased               3900 non-null  object  
 4   Category                     3900 non-null  object  
 5   Purchase Amount (USD)        3900 non-null  int64  
 6   Location                     3900 non-null  object  
 7   Size                         3900 non-null  object  
 8   Color                        3900 non-null  object  
 9   Season                       3900 non-null  object  
10  Review Rating                 3863 non-null  float64 
11  Subscription Status           3900 non-null  object  
12  Shipping Type                 3900 non-null  object  
13  Discount Applied              3900 non-null  object  
14  Promo Code Used               3900 non-null  object  
15  Previous Purchases            3900 non-null  int64  
16  Payment Method                3900 non-null  object  
17  Frequency of Purchases        3900 non-null  object  
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
[9]: df.describe(include = 'all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned **DataFrame** into the database for SQL analysis.

• SQL Business Analysis

SQL queries answered key business questions including revenue by gender, high-spending discount users, top-rated products, shipping comparisons, subscription spending, discount-dependent products, customer segmentation, top purchased products, repeat buyer behavior, and revenue by age groups.

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
Total rows: 839 Query complete 00:00:01		

3. Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	No.of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

- **Power BI Dashboard**

Dashboard includes Sales Overview, Customer Demographics, Product Performance, Subscription & Discount Analysis.



- **Business Recommendations**

1. Strengthen subscription benefits.
2. Build loyalty programs.
3. Target high-revenue age groups.
4. Optimize discount strategy.
5. Promote top-rated products.

❖ **Conclusion**

The project successfully integrates Python, SQL, and Power BI to deliver actionable retail insights. Demonstrates strong ETL, analytics, and BI capabilities.